SAT / MATH / FREE GUIDE

The Complete SAT Math Formula Guide

The essential formulas, what they mean, and how to put them to work.

By TestPrepUnraveled • Updated September 12, 2026 • 5 min read

Start with the relationship, not the formula

Before calculating, identify what the question gives you and what you need to find. Write down units. A formula is useful only when its variables match the quantities in the problem. The table below is a starter reference for frequently useful relationships, rather than an exhaustive list of everything that can appear.

Algebra: recognize the structure

In $y = mx + b$, m is the rate of change and b is the starting value. For a quadratic, factoring can reveal zeros quickly. If factoring is inconvenient, the quadratic formula works for any equation $ax^2 + bx + c = 0$ with $a \neq 0$. A negative discriminant means there are no real solutions.

Geometry and data

Draw and label a diagram, even if it is rough. Distinguish radius from diameter before using a circle formula. For data questions, the mean uses every value, while the median depends on order. An extreme value can change the mean substantially without changing the median.

Make your review active

Cover the formula column and rebuild each relationship from memory. Then solve one new problem. Keep a short error log with three entries: the relationship you missed, why you missed it, and a corrected example.

YOUR DESK-SIDE REFERENCE

Quick reference

Relationship	Formula	Use it when…
Slope	$m = (y_2 - y_1)/(x_2 - x_1)$	Two points; $x_2 \neq x_1$
Quadratic roots	$x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$	$ax^2 + bx + c = 0$
Circle	$A = \pi r^2$; $C = 2\pi r$	Area or circumference
Right triangle	$a^2 + b^2 = c^2$	c is the hypotenuse
Mean	sum of values ÷ number of values	Finding an average
Percent change	(new − old) ÷ old × 100%	Comparing to a nonzero starting value

Original worked example

A line passes through (2, 7) and (6, 19). What is its equation?

The slope is $(19 - 7)/(6 - 2) = 3$. Substitute (2, 7) into $y = 3x + b$: $7 = 6 + b$, so $b = 1$. The equation is $y = 3x + 1$.

A tip worth keeping

Use the available reference information thoughtfully. Knowing when a formula applies matters more than memorizing its appearance.

Your quick checklist

- Identify the exact task before working.
- Show the reasoning, not just the answer.
- Review the mistake and try a new example.

PUT IT INTO PRACTICE

Try it yourself

These original questions were written for this guide. Try each one before reading the answer key.

1. A circle has diameter 10. What is its area?

2. Solve $x^2 - 7x + 12 = 0$.

3. A price rises from $40 to $46. What is the percent increase?

Answers & explanations

1. 25π square units. The radius is 5, so $A = \pi \times 5^2$.

2. $x = 3$ or $x = 4$. Factor as $(x - 3)(x - 4) = 0$.

3. 15%. The increase is $6, and $6/40 \times 100\% = 15\%$.

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